SFEARNet: A Network Combining Semantic Flow and Edge-Aware Refinement for Highly Efficient Remote Sensing Image Change Detection

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***Abstract*— In change detection, the pseudovariations in the visual features of remote sensing (RS) images are attributed to imaging conditions, lighting, seasonal changes, atmospheric interference, and other factors. These pseudovariations yield a great challenge to change detection. The traditional change detection network usually suffers from upsampling information loss and blurred edges. Aiming at resolving the above problems, a semantic flow and edge-aware refinement network (SFEARNet) for highly efficient RS image change detection has been proposed. The pyramid feature enhancement module (PFEM) has been designed for the enhancement of differential information. The introduction of the semantic flow information transmission module (SFITM) enables the effective transmission and retaining of key information through semantic flow. An edge-aware refinement module (EARM) has been developed, designed to extract change edge and enhance the refinement effect of the edge. The experiments have been conducted on the LEVIR building change detection dataset (LEVIR-CD), WHU building dataset (WHU-CD), Google dataset (GZ-CD), and cropland change detection dataset (CLCD). In comparison with the existing methodologies, the experimental results demonstrate that SFEARNet attains the highest change detection accuracy and the smallest floating-point operations per second (FLOPs) while maintaining a similar number of parameters (Params). This enables more efficient change detection. In particular, the proposed method can effectively refine the edges of the change region, reduce the loss of upsampling information, and enhance differential feature extraction. This brings a new solution to the field of RS image change detection. The code is available at https://github.com/miao-0417/SFEARNet.**

***Index Terms*— Change detection, edge-aware, feature enhancement, remote sensing (RS), semantic flow.**

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I. Introduction

CHANGE detection is an active research field in remote sensing (RS). By comparing the same area two or more times, differences in ground objects can be monitored. The differences can be compared with the help of image-processing techniques and mathematical models. Change detection is used in many fields. These include land cover and land use (LCLU) change [1], [2], [3], [4] and disaster assessment [5], [6], [7], [8]. An important branch of change detection is bitemporal change detection. In comparison with the widely known RS image interpretation approaches such as image classification [9], [10], [11], [12], [13], [14], [15], [16], target detection [17], [18], [19], [20], [21], [22], [23], [24], [25], [26], [27], [28], [29], and super-resolution reconstruction [30], [31], [32], [33], change detection is based on the analysis of bitemporal images. There is a further need to improve bitemporal change detection techniques due to the increase in data and the increasing complexity of land cover. Two aspects in the following should not be ignored.

1) *Information Maintaining During Upsampling:* The network's performance has been greatly influenced by the high-resolution feature. Low-resolution features contain rich semantic information. During the upsampling process, this information may become blurred or lost. Reducing information loss during upsampling becomes particularly important [34], [35], [36]. This article introduces semantic flow for information transfer. This method can reduce information loss and obtain a strongly semantically representative feature map.
2) *Edge Information Utilization:* Edge information has a large impact on the accuracy. There are two main problems in current change detection based on edge information. The first one does not use real edge information for constraints [37], [38]. The second one uses real edge information, but the edge information shares a decoder with the change region information [39], [40], [41]. It is easy to cause incomplete extraction of the change area and create voids. A new decoder is proposed in this article. It uses the first and fourth layer features as a decoder for edge information. The preservation of the second and third layer features maintains integrity for change detection.

With these factors in mind, this article presents a network called semantic flow and edge-aware refinement network (SFEARNet). SFEARNet comprises semantic flow and edge-aware refinement. The network architecture used is the

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Siamese neural network. The SegFormer [42] structure is used for both the encoding and decoding parts of the network. Also, it builds on this foundation and includes improvements. First, a pyramid feature enhancement module (PFEM) is presented. This module can enhance the differential features. It integrates the original differential features, contextual features, and attention features. Moreover, a new module, known as the semantic flow information transmission module (SFITM), has been integrated into the decoder component of SegFormer to minimize the semantic information loss in the decoding stage. Therefore, to improve the edge awareness of the change detection network, an edge-aware refinement module (EARM) is introduced to focus more on the edges of the change region. This article makes the following contributions to knowledge.

1) *A Network Called SFEARNet Is Proposed in This Article:* The network integrates semantic flow and edge-aware refinement. Three modules are presented in this article, namely PFEM, SFITM, and EARM. With the same level of parameter count, SFEARNet not only ensures detection accuracy but also achieves the highest efficiency. Therefore, SFEARNet represents an appropriate solution in the domain of RS image change detection.
2) *This Article Makes Extensive Use of Multiscale Features:* In SFITM, it uses the transfer of semantic information between deep and shallow features to get high-resolution feature maps with high semantic information. In PFEM, contextual information can be captured at multiple scales using different dilation rates. PFEM can integrate original and enhanced features. The network is improved by the strategy of using multiscale features. There is also an increase in the accuracy and reliability of change detection.
3) *To Accurately Measure Change Boundaries, a New Decoder Component Is Introduced in EARM:* This design not only effectively separates edge information from change region information but also enhances the network's sensitivity and accuracy in identifying change boundaries. The module enables the capture of subtle changes in RS images with greater precision. This provides a solid foundation for the application in the field.

The remaining sections are organized as follows. Section II provides a brief overview of related work. Section III presents the details of SFEARNet. Section IV provides a comparison with the current state of the art. Section V discusses the methodology. Section VI summarizes this entire article.

II. Related Works

Deep learning excels at identifying sophisticated images. The technology has a high recognition rate. It also improves timeliness and automation. The convolutional neural network (CNN) is a pivotal deep learning algorithm. CNNs have wide applications in RS and are used in change detection. Advanced techniques such as feature enhancement, minimizing information loss, and the use of edge information are more often used. The area of RS image change detection has been expanded.

A. Deep Learning in RS

RS data typically include satellite images, radar data, and more. These data are large and complex. Traditional methods often struggle to fully exploit the potential information within it. With the introduction of deep learning, the processing and interpretation of this data have been greatly enhanced.

First, deep learning is used in the classification [43], [44], [45], [46] and segmentation [47], [48], [49], [50] of RS images. In classification, Ji et al. [43] proposed PASSNet. In segmentation, Chen et al. [47] proposed a coarse-to-fine task. Xu et al. [49] proposed RSSFormer. Second, deep learning is also widely used for target detection [26], [51], [52] and anomaly detection [23], [53], [54], [55]. A weakly supervised network was used by Han et al. [26]. Wang et al. [52] proposed FSoD-Net. Liu et al. [23] used an unsupervised anomaly network. Furthermore, deep learning is a crucial component in super-resolution reconstruction [56], [57], [58]. Xiao et al. [56] proposed EdiffSR. Last but not least, deep learning has demonstrated considerable potential in the domain of change detection [59], [60], [61]. Tang et al. [59] proposed WNet. Yuan et al. [61] proposed MFGFNet.

As deep learning algorithms improve, more and more uses will be found for it in RS. Especially in RS image change detection, deep learning will play its fast and efficient advantage to improve the efficiency.

B. Minimize Information Loss in Change Detection

One of the important issues in increasing accuracy is how to reduce information loss during the transmission process. Many scholars have put forward a variety of solutions to reduce the loss incurred during the transfer of information [34], [35], [36], [62], [63], [64]. Peng et al. [34] used a dense attention technique. This is achieved by multiple upsampling attention units. Fang et al. [35] used densely linked to reduce the question. Wang et al. [62] used distinct branches to combine feature maps of varied degrees. Gu et al. [63] combined the various levels of difference features through skip connection. Long et al. [36] proposed a high-level information refinement module. Wang et al. [64] introduced a structure with multiresolution parallel to fully utilize visual information at various resolutions.

Although considerable progress has been made in recent years in reducing information loss, current research is still largely focused on optimizing network structures to mitigate information loss during network transmission. However, another significant source of information loss has been overlooked. The source is the loss of details during the upsampling process. It is essential to implement more effective solutions. In this article, the method of semantic flow is employed to refine the transmission of information across images of different resolutions, resulting in highly abstracted high-resolution feature maps.

C. Feature Enhancement in Change Detection

Feature enhancement represents a pivotal aspect. This assists computers in enhancing the understanding of spatial and hierarchical characteristics. Also, feature strengthening can be defined as a way to enhance the performance of a model [65], [66], [67], [68], [69], [70], [71], [72], [73], [74], [75], [76]. Daudt et al. [65] proposed three topologies for a full CNN to extract deep information. Zhu et al. [66] added a feature

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comparison module (FCM). It is aimed at limiting channel numbers in feature fusion. A hybrid convolutional module was used by Wang et al. [67]. Li et al. [68] used multiscale convolution kernels. Features of ground object elements can be detailed using the different kernels. Chen et al. [69] enabled the efficient and effective modeling of situations within the spatial–temporal domain. Li et al. [70] learned the global context with transformer. Zhang and Shi [71] used a network with deep feature difference convolutional. Du et al. [72] proposed ESDSCNet with enhanced shallow feature difference and semantic context. Song et al. [73] incorporated local and global data from CNN and branch fusion in two directions. Wang et al. [75] proposed MTCNet. Bandara and Patel [76] proposed ChangeFormer. Liu et al. [74] proposed MSCANet. Both of them modeled long-range context information.

Considering feature enhancement is crucial when studying and analyzing areas of change. It can significantly improve the accuracy of identifying and understanding regional changes. Feature enhancement can highlight or amplify specific image or data features, making the change regions more pronounced. It becomes easier to detect and analyze. This article integrates the original differential features, differential contextual features, and differential attention features. This can improve the model's performance. Integrating these features helps understand complex regions better.

D. Edge Information in Change Detection

Edge information can be utilized as an auxiliary feature. Edge information is employed to identify changes in the boundaries and contours of objects. The edge can aid in the identification of the location and extent of changes in objects [37], [40], [41], [77], [78], [79]. Xiang et al. [40] used multimodal transformers that include edge guidance. Zhang et al. [77] introduced a module for preserving object integrity and compactness. Wang et al. [41] proposed a super-pixel module to create edges. Xia et al. [78] proposed an edge detection branch (EDB). Bai et al. [37] combined discriminative information and edge structure. Zhu et al. [79] introduced edge-aware and edge-guidance feature modules.

It is necessary to find change regions, and for this, accurate edge information is a basic requirement. The correct edge information can always describe and define the margins of the desired regions with great detail. Hence, effectively and efficiently obtaining the edge information has become vital for the change detection operation. In this article, the difference between the predicted and real edge information has been calculated through dice loss. It has helped enhance the extraction of edge information in change regions.

In summary, the enhancement of neural network technology in the recent past has seen the improvement of RS image change detection. Feature enhancement, reducing information loss, and incorporating edge information have been widely used and are still on the rise. These enhancements have not only refined the detection of changes but also increased the reliability and flexibility of the system. With the progress of technology, RS image change detection will go further to a more intelligent and precise level.

III. Methodology

This section is dedicated to the introduction of the methods utilized in the article. An initial summary of the combining semantic flow and edge-aware refinement network (SFRARNet) is provided. This is followed by a more detailed description of PFEM, SFITM, and EARM. Finally, the lose function is displayed.

A. Overall Structure of SFEARNet

SFEARNet is based on the SegFormer architecture. The achievement of this objective is made by means of PFEM, SFITM, and EARM. The overall network framework is a Siamese network. SFEARNet is illustrated in Fig. 1. The network uses SegFormer's encoder, the MixTransformer. The MixTransformer is the core feature extraction module.

Initially, two RS images of an area, captured at disparate points in time, are fed into the shared-weight encoder. This process yields feature vectors, designated as F_i and f_i, where i is the level of the feature vector. Subsequently, feature vector pairs (F_i, f_i), derived from different RS images and at the same level, are transmitted to PFEM. Following PFEM processing, an enhanced feature representation, designated $F_{\mathrm{PFEM}i}$, is generated.

Subsequently, EARM is applied, comprising two stages. In stage one, the most representative feature scales F_{PFEM1} and F_{PFEM4} are selected. They are inputs for subsequent processing within EARM. This module employs a distinctive edge-awareness mechanism to refine edge information. This facilitates the network's ability to discern a more accurate edge. The output edges generated by EARM are evaluated with the ground-truth edges through the calculation of loss functions. This process serves to enhance the capability of edge recognition while also improving the accuracy of edge information extraction. In stage two, the edge information generated by EARM acts on $F_{\mathrm{PFEM}i}$. Here, the resolution of edge information is adjusted through max pooling to refine $F_{\mathrm{PFEM}i}$ and obtain a more refined feature representation, $F_{\mathrm{EARM}i}$.

Finally, the refined features $F_{\mathrm{EARM}i}$ are transmitted to the SegFormer decoder. To improve the precision of generated images, the conventional bilinear interpolation upsampling technique employed within the SegFormer framework has been superseded by SFITM. SFITM module enhances the generation of large-scale RS images, thus optimizing the network's overall performance.

B. Semantic Flow Information Transmission Module

It is very important to get feature maps with high resolution and strong semantic representation for improving performance. The upsampling module plays an important role in change detection. Traditional upsampling methods, such as bilinear interpolation, can generate relatively smooth results in the process of processing. It is inevitable that such methods will result in some details getting lost. Inspired by the literature [80], this article proposes an SFITM, as shown in Fig. 2.

This module preserves the details of the original images well. It maximizes the retention and transmission of key

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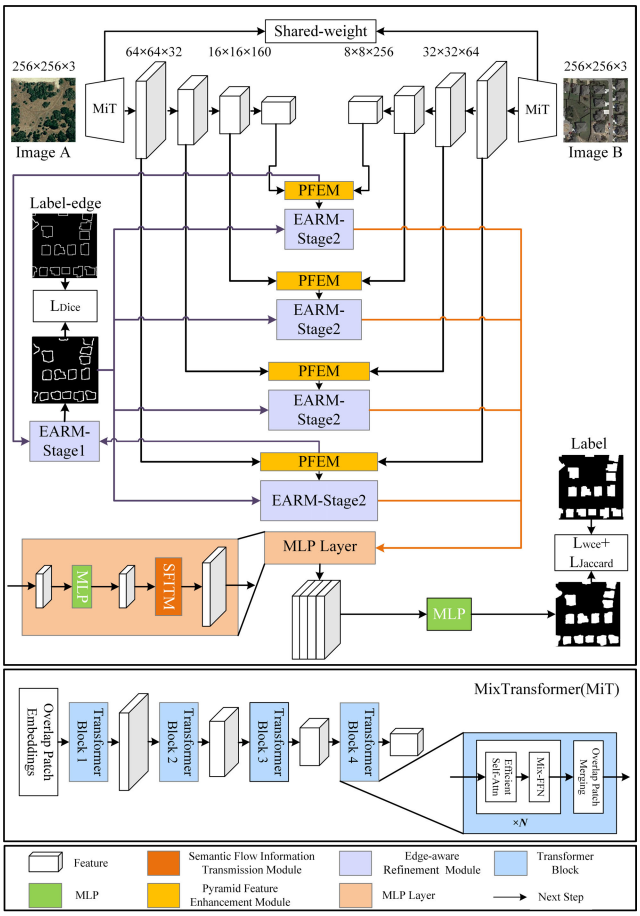


Fig. 1. Framework of SFEARNet.

details within the images. A multiscale feature integration strategy is also employed. This strategy makes full use of information from feature maps with different resolutions. The core of this module lies in the information transmission based on semantic flow. The semantic representativeness of deep features is preserved. This makes it possible to obtain high-resolution features with strong semantic representation. Semantic information is transmitted between different resolutions. The integration of multiscale features has been achieved.

First, given two levels of input feature maps $F_l^{H_l \times W_l \times C_l}$(deep feature) and $F_p^{H_p \times W_p \times C_p}$(shallow feature), the resolution of $F_l^{H_p \times W_p \times C_p}$ is equal to that of $F_p^{H_p \times W_p \times C_p}$ by bilinear upsampling, as shown in (1). Two feature maps are concatenated. Then, they are convolved with a kernel size of 3×3 with two filters. A semantic flow field Δp is predicted, as shown in (2). The size of the semantic flow field is $H_p \times W_p \times 2$. Each element in the flow field corresponds to an offset, which represents the positional offset of each position in the previous level in the current level, and the formulas are given as follows:

$$F_l^{H_p \times W_p \times C_p} = \text{Bilinear Upsample}\left(F_l^{H_l \times W_l \times C_l}\right) \tag{1}$$

$$\Delta p = \text{Conv}_{3\times 3}\left(\text{Concat}\left(F_l^{H_p \times W_p \times C_p}, F_p^{H_p \times W_p \times C_p}\right)\right) \tag{2}$$

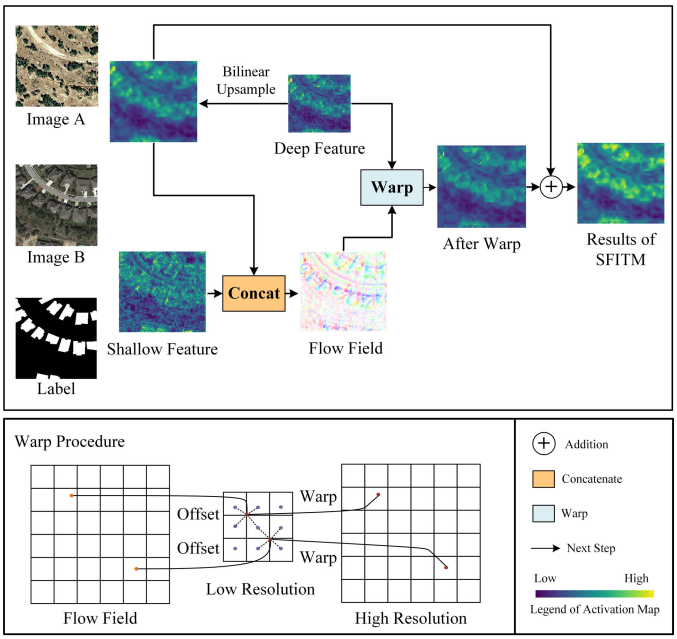


Fig. 2. Framework of SFITM.

where H denotes the length, W denotes the width, C denotes the number of channels, and $\text{Conv}_{3\times 3}$ represents 3×3 convolutions.

Next, after obtaining the semantic flow field, each position f_p of the previous level is mapped to a position f_l of the current level, as shown in (3). After mapping, linear interpolation of the current level feature map is used to obtain the final feature map F. This process uses a differentiable bilinear sampling mechanism, as shown in (4). By adding it with the results of F_l bilinear upsampling above, the output F_{SFITM} of the module is obtained, as shown in (5). The formulas are given as follows:

$$f_l = f_p + \frac{\Delta p(f_p)}{2} \tag{3}$$

$$F = \sum_{f' \in N_{(fl)}} \omega'_f F_l(f'_l) \tag{4}$$

$$F_{\text{SFITM}} = F + \text{Bilinear Upsample}\left(F_l^{H_l \times W_l \times C_l}\right) \tag{5}$$

where ω'_f denotes the bilinear interpolation weights and $N_{(fl)}$ represents neighbors of the warped points f_l in F_l

In this article, SFITM is utilized to replace the traditional bilinear interpolation upsampling method. The warp operation can perform geometric transformations on images. It can spatially align an image with another image taken at a different point in time. Bilinear interpolation is an image resampling technique. It is used to adjust the image resolution. By combining the warp operation with bilinear interpolation, the advantages of both processing methods can be integrated. SFITM preserves the spatial alignment information of the image. It also maintains a high resolution. This aids in subsequent change detection tasks. SFITM allows for a more accurate restoration of image detail. This helps to improve the model's perception. Ultimately, it improves the accuracy and precision of change detection.

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C. Pyramid Feature Enhancement Module

Accurately measuring the differences between bitemporal features is crucial. Traditional methods often rely on simple difference calculations. This leads to the extraction of ambiguous differences in features. The contextual information associated with the difference features is not sufficiently preserved. The PFEM, as shown in Fig. 3, is proposed to overcome this limitation. The module uses a multiscale fusion strategy. First, it preserves the original differential features. Next, dilated convolution is introduced in the feature enhancement stage, allowing contextual data to be processed at various scales using different dilation rates. Finally, an attention mechanism is used to enhance features from two branches. The module integrates the original differential features, contextual features, and attention features.

The receptive fields of dilated convolutions with different dilation rates exhibit a pyramid-like structure. Therefore, this module is aptly named the PFEM, reflecting its unique advantages in multiscale feature extraction. Through precise operations, this module not only preserves the integrity of the original features but also accurately extracts and significantly enhances the differential features. This ensures the effective utilization and accurate analysis of differential features in the change detection process. The process is given as follows.

On the one hand, on the bitemporal features F_c and F_c, where c is the channel number, a difference calculation is performed. This method captures the differences between the feature values of bitemporal images. This study draws inspiration from the ASPP module in DeeplabV3+ [81]. A multiscale dilated convolution method is hereby introduced with a view to capturing contextual information from changing regions at different scales. This article uses the 1×1 convolution for original feature extraction, resulting in $F_{1\times1}$ as shown in (6). In addition, four parallel dilated 3×3 convolutions with different dilation rates ($d = 1, 2, 3, 4$) are used. Contextual features can be extracted from the different dilation rates. This method results in $F_{3\times3}^{d,c}$, as shown in (7). This approach maintains the spatial integrity of the convolutional features while capturing regions of varying sizes using different receptive fields. Subsequently, the multiscale contextual features are concatenated, and 1×1 convolution is employed to downsample the channel, resulting in F_{cat}, as shown in (8). Then, add F_{cat} to the original difference features, resulting in F_{add}, as shown in (9). This process retains the original features with richer context information, enhancing the features effectively. The formulas are given as follows:

$$F_{1\times1} = \text{Conv}_{1\times1}(\text{abs}(F_C - f_C)) \tag{6}$$

$$F_{3\times3}^{d,c} = \text{Conv}_{3\times3}^{d,c}(\text{abs}(F_C - f_C)), \quad d \in 1, 2, 4, 8 \tag{7}$$

$$F_{\text{cat}} = \text{Conv}_{1\times1}\left(\text{Concat}\left(F_{1\times1}, F_{3\times3}^{d,c}\right)\right), \quad d \in 1, 2, 4, 8 \tag{8}$$

$$F_{\text{add}} = F_{\text{cat}} + \text{abs}(F_C - f_C) \tag{9}$$

where d is the dilation rate, c is the number of channels, $\text{Conv}_{1\times1}$ represents 1×1 convolution, and abs represent the absolute. $\text{Conv}_{3\times3}^{d,c}$ represents 3×3 convolutions with the dilation rates and channel numbers. Concat represents concatenation.

On the other hand, the features from these two phases are concatenated. The information in the bitemporal features (F_c, f_c) can be fully exploited. The feature information of the original bitemporal image is also retained. This concatenation results in $F_{\text{cat}}^{2\times c}$, as shown in (10). $F_{\text{cat}}^{2\times c}$ is sent to the channel and spatial attention. The attention is named convolutional block attention module (CBAM) [82]. CBAM enables the model to perceive key information with greater efficacy. It is worth noting that CBAM can significantly improve network performance without increasing network complexity. After the feature is processed by CBAM, 1×1 convolution is used for channel downsampling to optimize the feature expression resulting in F_{conv}, as shown in (11). Subsequently, this article adds the downsampled features to F_{add}. F_{add} represents the different features processed and enhanced by multiscale dilated convolution. This addition results in F_{PFEM}, as shown in (12). The objective is to integrate the different features from the original image features. It can use the original image's feature information. This process can improve SFEARNet's change accuracy. The formulas are given as follows:

$$F_{\text{cat}}^{2\times c} = \text{Concat}(F_c, f_c) \tag{10}$$

$$F_{\text{conv}} = \text{Conv}_{1\times1}\left(\text{CBAM}\left(F_{\text{cat}}^{2\times c}\right)\right) \tag{11}$$

$$F_{\text{PFEM}} = F_{\text{conv}} + F_{\text{add}} \tag{12}$$

where CBAM stands for the convolutional block attention module. c is the number of channels.

Differential features are enhanced using PFEM. PFEM uses a multiscale pyramid structure. The use of dilation convolutions with different dilation rates is a necessary step. The different information of change regions can be captured and enhanced. At the same time, the original features of the bitemporal images are concatenated. Then, the CBAM attention mechanism is applied. The enhanced change detection features are obtained by adding the features obtained from these two methods. The module combines the original differential, contextual, and attention features. PFEM enhances the model's detection accuracy.

D. Edge-Aware Refinement Module

Edges are fundamental visual elements in images, containing object shapes and contours. The incorporation of edge features is a pivotal aspect of deep learning visual models, exerting a considerable influence on image comprehension and model efficacy. In designing and training deep learning models, efficiently and accurately extracting and utilizing edge information are crucial. Inspired by the existing literature [83], this article proposes an EARM, as shown in Fig. 4. This module aims to extract and optimize edges of the changing region, using this edge information to refine network features.

In this module, a decoder is proposed. It is used to identify and precisely extract edge information of the changing regions in the image. The decoder primarily utilizes shallow and deep features from the network structure. This allows decoders to take advantage of both shallow features in terms of detail and deep features in terms of semantics. It can locate the edges of

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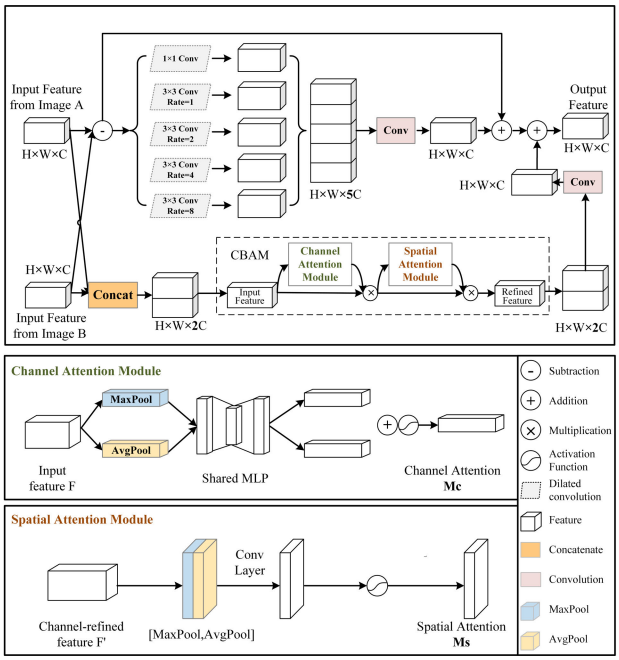


Fig. 3. Framework of PFEM.

changing regions. This decoder overcomes the limitations of single feature layers in edge detection by combining features from different layers. It achieves higher accuracy in complex change detection.

Edge information is extracted using the Sobel edge detection operator. The calculation process is shown in (13)–(17). This is followed by threshold segmentation to extract the edges of changed objects, with the threshold value set to 20, as shown in (18). This module is divided into two stages. Stage 1 involves the extraction and optimization of edge information. Stage 2 uses this edge information to refine the features

$$S_x(i,j) = \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix} \tag{13}$$

$$G_x = \sum_{i=-1}^{1} \sum_{j=-1}^{1} I(x+i, y+j) \cdot S_x(i,j) \tag{14}$$

$$S_y(i,j) = \begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix} \tag{15}$$

$$G_y = \sum_{i=-1}^{1} \sum_{j=-1}^{1} I(x+i, y+j) \cdot S_y(i,j) \tag{16}$$

$$G = \sqrt{G_x^2 + G_y^2} \tag{17}$$

where $S_x(i,j)$ denotes the horizontal direction gradient operator. $S_y(i,j)$ is the operator for the vertical gradient. $I(x,y)$ is the pixel value of the image at the location (x,y). G_x is the horizontal gradient. G_y is the vertical gradient. G is the magnitude of the gradient

$$\text{output}(x,y) = \begin{cases} 255, & \text{if input}(x,y) > 20 \\ 0, & \text{if input}(x,y) \le 20 \end{cases} \tag{18}$$

where input(x,y) is the pixel value.

In Stage 1, the shallow and deep features $F_{\text{PFEM}}^{H_s \times W_s \times C_s}$ and $F_{\text{PFEM}}^{H_d \times W_d \times C_d}$ are enhanced by PFEM. Perform a 3×3 convolution on deep features and shallow features to obtain $F_{\text{conv}}^{H_s \times W_s \times C_s}$ and $F_{\text{conv}}^{H_d \times W_d \times C_d}$, as shown in (19) and (20). Shallow features are effective at capturing detail information in images, such as edges. Deep features can capture high-level semantic information, such as categories. The shallow features here use the first-layer features. The deep features here use the fourth-layer features. This enables the concurrent utilization of the edge information contained within first-layer features and the category information embedded within fourth-layer features. Use SFITM to adjust the resolution of deep features to match that of shallow features. Concatenate the deep features after SFITM with the shallow features to obtain $F_{\text{cat}}^{H_s \times W_s \times C_s}$, as shown in (21), to create a rich feature representation. Double convolution is performed on the concatenated features to further extract and emphasize the edge features of the changing region, resulting in $F_{\text{edge}}^{H_s \times W_s \times C_s}$, as shown in (22). This series of fine operations aims to accurately identify the changing regions in the image and assign higher weights to their edge information. The corresponding formulas are given as follows:

$$F_{\text{conv}}^{H_s \times W_s \times C_s} = \text{Conv}_{3\times3}\left(F_{\text{PFEM}}^{H_s \times W_s \times C_s}\right) \tag{19}$$

$$F_{\text{conv}}^{H_d \times W_d \times C_d} = \text{Conv}_{3\times3}\left(F_{\text{PFEM}}^{H_d \times W_d \times C_d}\right) \tag{20}$$

$$F_{\text{cat}}^{H_s \times W_s \times C_s} = \text{Concat}\left(F_{\text{conv}}^{H_s \times W_s \times C_s}, \text{SFITM}\left(F_{\text{conv}}^{H_d \times W_d \times C_d}\right)\right) \tag{21}$$

$$F_{\text{edge}}^{H_s \times W_s \times C_s} = \text{Conv}_{3\times3}\left(\text{Conv}_{3\times3}\left(F_{\text{cat}}^{H_s \times W_s \times C_s}\right)\right) \tag{22}$$

where H stands for the length, W stands for the width, C stands for the number of channels, and SFITM represents the semantic flow information transmission module.

In Stage 2, the edge information is used to refine the output features of PFEM. On the abovementioned edge information, a maximum-pooling operation is performed. It results in $F_{\text{edge-feature}}^{H \times W \times C}$, as shown in (23). Maximum pooling can extract edge features that are most significant. The feature values output from PFEM are multiplied by this pooled edge information on an element-by-element basis. The representation of the edge is directly improved by this procedure. The result of the multiplication is added to the values of the original characteristics. It is able to preserve the nonedge information in the original features. The method gets $F_{\text{edge-correction}}^{H \times W \times C}$, as shown in (24). This step ensures that the edge information is effectively enhanced, while other important information in the original features is also preserved. Finally, the edge-enhanced features are passed into the CBAM attention mechanism. CBAM can further focus on the key channels and spatial positions in the feature map, thereby obtaining the features after edge refinement resulting in F_{ESGM}, as shown in (25). These feature values not only contain rich edge information but also enhance key regions in the image through the attention mechanism. This results in a more accurate and more robust representation of the features. The specific calculation formulas are given as follows:

$$F_{\text{edge-feature}}^{H \times W \times C} = \text{MaxPool}\left(F_{\text{edge}}^{H_l \times W_l \times C_l}\right) \tag{23}$$

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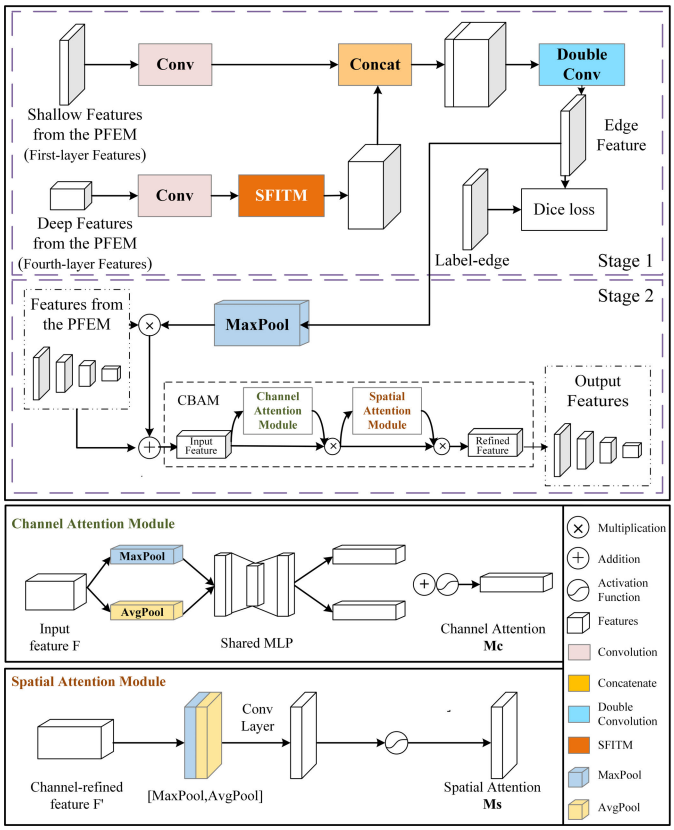


Fig. 4. Framework of EARM.

$$F_{\text{edge-refine}}^{H\times W\times C} = F_{\text{edge-feature}}^{H\times W\times C} \times F_{\text{PFEM}}^{H\times W\times C} + F_{\text{PFEM}}^{H\times W\times C} \tag{24}$$

$$F_{\text{ESGM}} = \text{CBAM}\left(F_{\text{edge-refine}}^{H\times W\times C}\right) \tag{25}$$

where H denotes the length, W denotes the width, and C denotes the number of channels. MaxPool represents the max pooling.

This article proposes an EARM that utilizes both shallow and deep features to accurately extract edge information. More importantly, this edge information is applied to the feature values at different stages to enhance their refinement. This strategy is particularly crucial in change detection tasks, producing more accurate and detailed change detection outcomes.

E. Loss Function

EARM is crucial to the network structure. The accuracy of the edges is directly influenced by this factor. This, in turn, has a profound impact on the subsequent extraction of deep change features. Dice loss is used to ensure that the training phase produces precise boundary prediction information by the use of original edge data.

The cross-entropy (CE) and Jaccard losses are calculated for the label predictions and the ground-truth values. This evaluation assesses the model's performance in predicting categories.

In conclusion, the total loss is a combination of CE, Jaccard, and dice losses. This is designed to improve the accuracy of detecting edges and predicting categories. The formulas are given as follows:

$$L_{\text{ce}} = -\frac{1}{N}\sum_{n=0}^{N-1} y_n \log(x_n) \tag{26}$$

$$L_{\text{Jaccard}} = 1 - \frac{|A \cap B|}{|A \cup B|} \tag{27}$$

$$L_{\text{Dice}} = 1 - \frac{2\times|A \cap B|}{|A| + |B|} \tag{28}$$

$$L_{\text{total}} = L(\text{GT}_{\text{label}}, \text{pre}_{\text{label}}) + L(\text{GT}_{\text{label}}, \text{pre}_{\text{label}}) + L(\text{GT}_{\text{edge}_{\text{label}}}, \text{pre}_{\text{edge}}) \tag{29}$$

where L_{ce} denotes the CE loss. L_{Jaccard} denotes the Jaccard loss. L_{Dice} denotes the dice loss. x is the input value. y is the label value. N is the number of categories. $|A|$ and $|B|$ indicate the numbers of elements A and B. GT represents the ground truth, and Pre represents predictions.

IV. Experiments and Results

The SFEARNet is compared with the most advanced network models in detail on four datasets. This series of experiments has been designed to verify the advantages and effectiveness of SFEARNet.

A. Datasets

LEVIR building change detection dataset (LEVIR-CD) [84], WHU building dataset (WHU-CD) [85], Google dataset (GZ-CD) [86], and cropland change detection dataset (CLCD) [74] datasets are used for the experiments.

1) *LEVIR-CD [84]:* The dataset was acquired from Google Earth. It has a resolution of 0.5 m. The image size is 1024 × 1024. Notable changes have been observed in the land use patterns, particularly within the construction industry. The types of changes cover various types of buildings. In this experiment, the size is reduced to 256 × 256, with a nonoverlapping cut. The dataset is divided into a training set/validation set/test set, with a size of 7120/1024/2048.
2) *WHU-CD [85]:* A pair of bitemporal aerial photographs comprise the dataset. The resolution is 0.2 m. The size is 32 507 × 15 354. These images cover areas where earthquakes have occurred and have been reconstructed for many years, with the majority of the data about the reconstruction of buildings. The dataset is cut to 256 × 256 with nonoverlapping. The training set, the validation set, and the test set are divided according to a ratio of 8:1:1, with a size of 6096/762/762.
3) *GZ-CD [86]:* The data collection period is 2006–2019. The spatial resolution is 0.55 m. The size range is from 1006 × 1168 to 4936 × 5224 pixels. The image has been cut into nonoverlapping 256 × 256 pixel squares. The training set, the validation set, and the test set were divided according to a ratio of 8:1:1, with a size of 2882/361/360.
4) *CLCD [74]:* These data were collected in 2017 and 2019. The data are from Gaofen-2. Spatial resolution ranges from 0.5 to 2 m. A 512 × 512 bitemporal image and a binary label are used for each sample set. In total, there are 600 pairs of variation images. The CLCD's

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labeled types mainly include cropland change. In the application, it is cropped to 256 × 256 size. The total size of the training set, the validation set, and the test set is calculated to be 1440/480/480.

B. Evaluation Indicators

To verify the proposed network performance, this article uses six indicators to evaluate the similarity between the predicted results and the real values. These are precision (Pre), recall (Rec), $F1$ score ($F1$), intersection over union (IOU), overall accuracy (OA), and kappa coefficient (Kappa). The indicators are defined as follows:

$$\mathrm{OA} = \frac{\mathrm{TP}+\mathrm{TN}}{\mathrm{TP}+\mathrm{TN}+\mathrm{FP}+\mathrm{FN}} \tag{30}$$

$$\mathrm{Pre} = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FP}} \tag{31}$$

$$\mathrm{Rec} = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FN}} \tag{32}$$

$$\mathrm{F1} = \frac{2\times \mathrm{Pre}\times \mathrm{Rec}}{\mathrm{Pre}+\mathrm{Rec}} \tag{33}$$

$$\mathrm{IOU} = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FP}+\mathrm{FN}} \tag{34}$$

$$pe = \frac{(\mathrm{TP}+\mathrm{FP})\times(\mathrm{TP}+\mathrm{FN})}{(\mathrm{TP}+\mathrm{TN}+\mathrm{FP}+\mathrm{FN})^2} + \frac{(\mathrm{FN}+\mathrm{TN})\times(\mathrm{FP}+\mathrm{TN})}{(\mathrm{TP}+\mathrm{TN}+\mathrm{FP}+\mathrm{FN})^2} \tag{35}$$

$$\mathrm{Kappa} = \frac{\mathrm{OA}-\mathrm{pe}}{1-\mathrm{pe}} \tag{36}$$

where TP, FP, TN, and FN are the number of true positive, false positive, true negative, and false negative samples.

C. Comparative Experiment

SFEARNet is compared with several state-of-the-art bitemporal change detection networks. These include pure CNN networks, attention-based networks, CNN-transformer networks, pure transformer networks, and edge-based networks. Pure CNN models include fully convolutional early fusion (FC-EF) [65], FC-Siam-conc [65], and FC-Siam-diff [65]. Attention-based networks include STANet [84], difference enhancement and spatial–spectral nonlocal (DESSN) [87], and efficient ultralightweight spatial–spectral feature cooperation (USSFC) [88]. CNN-transformer networks include bitemporal image transformer (BIT) [69], MSCANet [74], and AMTNet [89]. Pure transformer network includes ChangeFormer [76]. Edge-based network includes EATDer [39]. Through a comparison with these methods, the advantages and the effectiveness of SFEARNet can be evaluated in a more precise way.

1) *FC-EF [65]:* The network used an early fusion category. The images from different phases are superimposed on the channel. The superimposed image is sent to the network for training and learning differences.
2) *FC-Siam-Conc [65]:* This is an intuitive way. The network uses a Siamese structure. During the decoding steps, the features are concatenated in two skip connections.
3) *FC-Siam-Diff [65]:* To be different with FC-Siam-conc, the absolute value of two different images has been concatenated.
4) *STANet [84]:* The study is based on a spatial–temporal attention module. The spatial–temporal attention module consists of the basic spatial–temporal attention module and the pyramid spatial–temporal attention module.
5) *BIT [69]:* The network proposed a bitemporal image transformer. It expressed images in a few tokens. Finally, it used a transformer encoder.
6) *ChangeFormer [76]:* The transformer-based network and the unified transformer encoder with multilayer perception decoder.
7) *MSCANet [74]:* It is a CNN-transformer network. The network utilized a CNN-based feature extractor and a transformer-based multiscale context aggregation. In addition, it employed a multibranch prediction head.
8) *DESSN [87]:* The network is based on difference enhancement and spatial–spectral nonlocal. The network is comprised of three distinct modules. These modules are the difference enhancement module, the spatial–spectral nonlocal module, and the asymmetric double convolution with ghost module.
9) *AMTNet [89]:* An attention-based multiscale transformer network using a CNN transformer structure. To model contextual information, the network uses attention and transformer models. The network also uses feature sharing to bridge the domain gap.
10) *USSFC [88]:* An efficient ultralightweight spatial–spectral feature cooperation network. The network used multiscale decoupled convolution and spatial–spectral feature cooperation strategies.
11) *EATDer [39]:* EATDer named edge-assisted adaptive transformer detector. It is a Siamese network with an edge-aware decoder. In each Siamese branch, it has three self-adapting vision transformer blocks. Full-range fusion modules connect two branches.

On each dataset, this article uses official code to reproduce as much as possible. In order to make the analysis more comprehensive, this article evaluates the performance of SFEARNet in SegFormer encoder, MixTransformer-b0.

D. Implementation Details

The proposed SFEARNet and all baseline models are implemented using PyTorch in this article. Training is performed on an NVIDIA GeForce ray tracing texel extreme (RTX) 3090 graphics processing unit (GPU). The input size was set to 256 × 256 pixels. The batch size was 8. The initial learning rate was set to $1e^{-4}$. The ReduceLROnPlateau strategy was employed to adjust the learning rate, having a factor of 0.5 and patience of 6 epochs. Model parameters (Params) were optimized using the AdamW optimizer with a weight decay of $1e^{-3}$. Various methods were applied to the input data during training, including random rotation, horizontal flipping, vertical flipping, and bitemporal phase exchange. All the methods are data augmentation operations. The SFEARNet model was trained for 100 epochs. During the testing phase, no data augmentation techniques were used, and only test data were utilized for evaluation.

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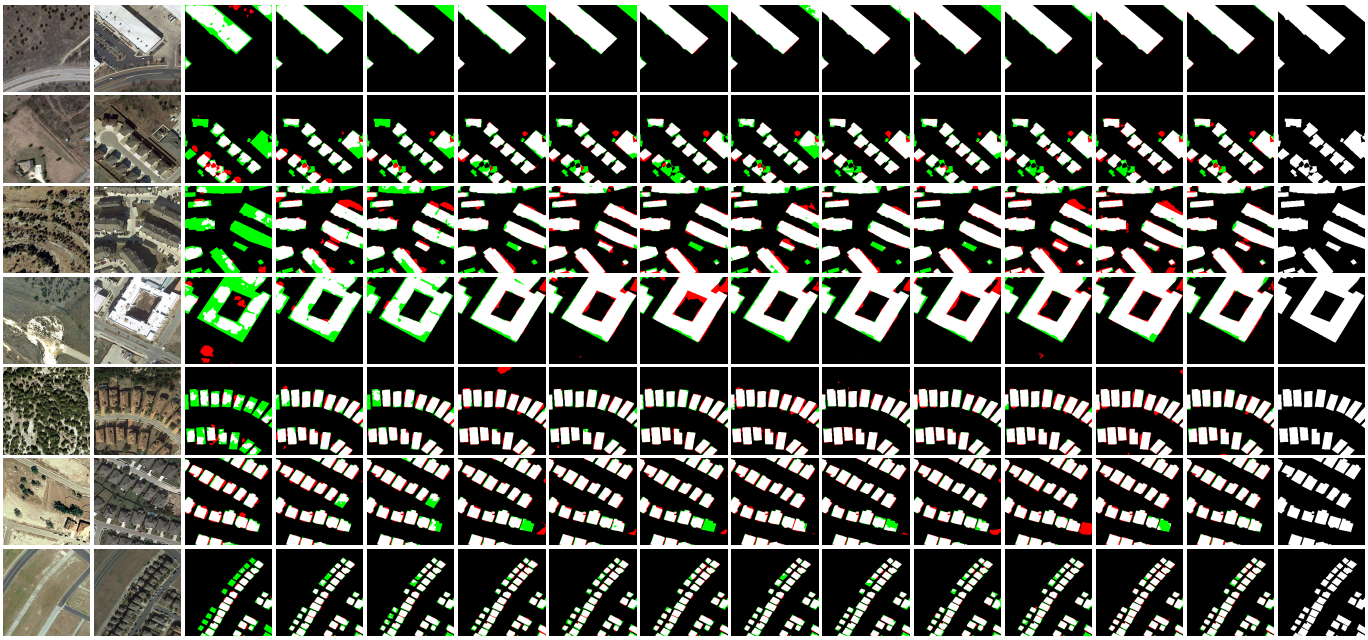


Fig. 5. Visualization comparisons on the LEVIR-CD dataset. (a) T1 image. (b) T2 image. (c) FC-EF. (d) FC-Siam-Conc. (e) FC-Siam-Diff. (f) STANet. (g) BIT. (h) MSCANet. (i) DESSN. (j) ChangeFormer. (k) AMTNet. (l) USSFC. (m) EATDer. (n) SFEARNet. (o) Ground truth. White indicates true positive. Black indicates true negative. Red indicates false positive. Green indicates false negative.

E. Experimental Results

The accuracy of LEVIR-CD, WHU-CD, GZ-CD, and CLCD is compared with advanced methods. It can assess the effectiveness of the proposed SFEARNet, as illustrated in Tables I–IV. This article presents SFEARNet predictions and existing methods visually. It can be seen in Figs. 5–8. The diagrams offer a direct representation of the method's performance.

Compared to several existing methods, SFEARNet shows a good performance on the LEVIR-CD dataset. This is shown in Table I. In four key metrics, OA, $F1$, IOU, and Kappa, SFEARNet achieves the highest scores. SFEARNet improves IOU by 0.57, 1.87, 1.55, and 2.07 percentage points relative to AMTNet, ChangeFormer, EATDer, and BIT. Meanwhile, SFEARNet improves $F1$ by 0.93, 0.34, and 1.12 percentage points compared to EATDer, AMTNet, and ChangeFormer. In addition, a visual comparison between SFEARNet and other methods is presented, as shown in Fig. 5. When dealing with changing objects of different sizes, SFEARNet shows excellent performance. SFEARNet is particularly sensitive to object edges when detecting large changes and can accurately capture boundary information. In addition, SFEARNet effectively detects changes in dense roofs. Compared to other methods, SFEARNet not only captures the general outline of small buildings during change detection but also retains more complete detailed information. The result is a more precise delineation of the areas of change.

SFEARNet is compared with other methods on the WHU-CD dataset. It can be seen in Table II. SFEARNet performs best in the five main metrics. These are Rec, OA, $F1$, IOU, and Kappa. Specifically, SFEARNet improves IOU by 0.72, 2.92, and 3.01 percentage points relative to AMTNet, BIT, and MSCANet. Meanwhile, SFEARNet achieves 2.19, 7.27, and 8.72 percentage points' improvement in IOU compared to BIT, USSFC, and EATDer. Meanwhile, SFEARNet achieves 0.42, 1.72, and 4.84 percentage points' improvement in $F1$ compared to AMTNet, BIT, and USSFC. A visual comparison between SFEARNet and other methods on the WHU-CD is shown in Fig. 6. SFEARNet achieves satisfactory results in large change regions, irregular change regions, and small to medium change regions. This result further validates the superior performance of SFEARNet.

TABLE I

QUANTITATIVE RESULTS (%) FOR THE LEVIR-CD DATASET. BOLD VALUES ARE BEST

Methods	Year	OA	Pre	Rec	F1	IOU	Kappa
FC-EF	2018	97.70	82.83	69.19	75.40	60.51	74.20
FC-Siam-Conc	2018	98.61	88.08	84.17	86.08	75.57	85.35
FC-Siam-Diff	2018	98.63	90.40	81.72	85.84	75.20	85.12
STANet	2020	98.92	90.75	87.68	89.19	80.49	88.62
BIT	2021	98.96	91.32	87.94	89.60	81.16	89.06
MSCANet	2022	98.97	91.27	88.24	89.73	81.37	89.19
DESSN	2022	98.89	90.58	87.39	88.96	80.11	88.37
ChangeFormer	2022	98.97	91.22	88.28	89.73	81.36	89.18
AMTNet	2023	99.05	**91.71**	89.33	90.50	82.66	90.00
USSFC	2023	98.88	90.31	87.38	88.82	79.89	88.23
EATDer	2024	98.95	88.13	**91.77**	89.91	81.68	89.36
SFEARNet		**99.07**	91.43	90.27	**90.85**	**83.23**	**90.36**

TABLE II

QUANTITATIVE RESULTS (%) FOR THE WHU-CD DATASET. BOLD VALUES ARE BEST

Methods	Year	OA	Pre	Rec	F1	IOU	Kappa
FC-EF	2018	97.99	79.49	69.68	74.26	59.06	73.22
FC-Siam-Conc	2018	97.43	69.25	68.88	69.06	52.74	67.72
FC-Siam-Diff	2018	98.12	80.02	73.04	76.37	61.78	75.40
STANet	2020	99.05	93.83	82.59	87.86	78.34	87.36
BIT	2021	99.25	94.75	86.89	90.65	82.89	90.26
MSCANet	2022	99.26	**95.79**	85.93	90.59	82.80	90.21
DESSN	2022	98.92	88.82	84.72	86.72	76.55	86.16
ChangeFormer	2022	98.73	89.16	79.01	83.78	72.09	83.12
AMTNet	2023	99.34	93.93	90.04	91.94	85.09	91.60
USSFC	2023	98.99	90.33	84.88	87.52	77.81	87.00
EATDer	2024	98.88	86.38	86.82	86.60	76.36	86.01
SFEARNet		**99.38**	94.25	**90.55**	**92.36**	**85.81**	**92.04**

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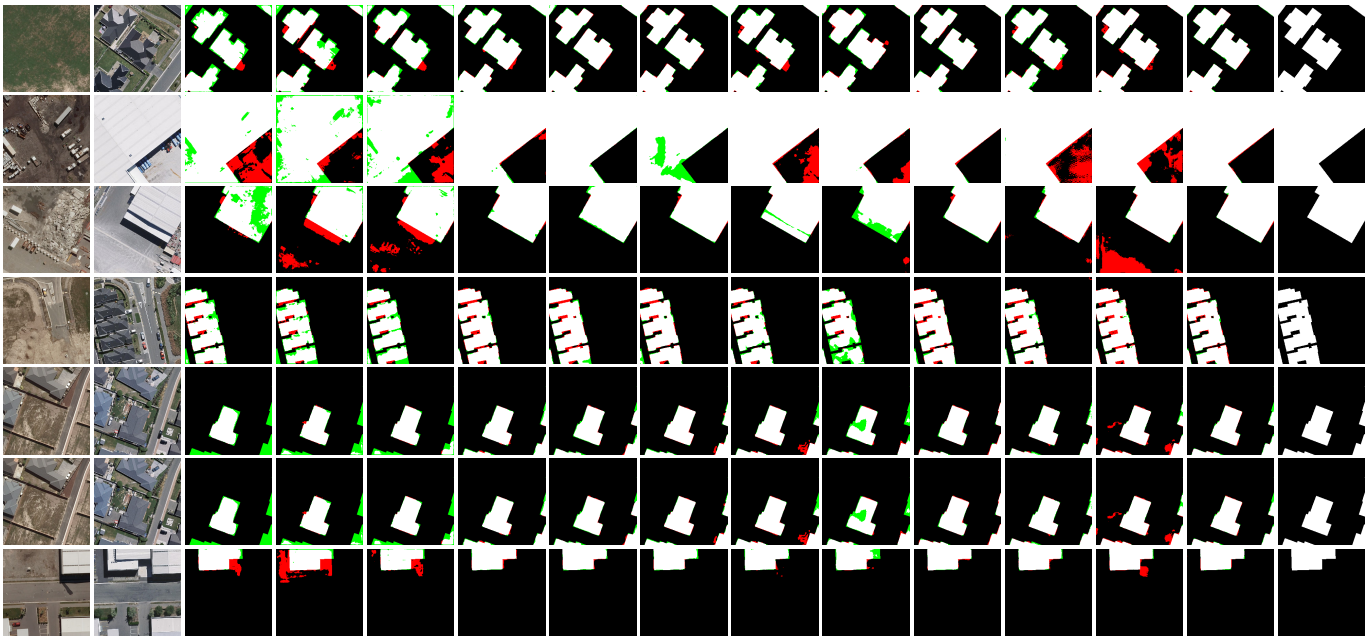


Fig. 6. Visualization comparisons on the WHU-CD dataset. (a) T1 image. (b) T2 image. (c) FC-EF. (d) FC-Siam-Conc. (e) FC-Siam-Diff. (f) STANet. (g) BIT. (h) MSCANet. (i) DESSN. (j) ChangeFormer. (k) AMTNet. (l) USSFC. (m) EATDer. (n) SFEARNet. (o) Ground truth. White indicates true positive. Black indicates true negative. Red indicates false positive. Green indicates false negative.

TABLE III
QUANTITATIVE RESULTS (%) FOR THE GZ-CD DATASET. BOLD VALUES ARE BEST

Methods	Year	OA	Pre	Rec	F1	IOU	Kappa
FC-EF	2018	94.45	73.82	59.14	65.67	48.89	62.69
FC-Siam-Conc	2018	94.25	73.47	56.20	63.69	46.72	60.62
FC-Siam-Diff	2018	93.80	70.34	53.43	60.73	43.60	57.43
STANet	2020	97.27	88.96	79.40	83.91	72.28	82.42
BIT	2021	96.81	83.46	80.35	81.87	69.31	80.12
MSCANet	2022	96.91	87.64	76.34	81.60	68.92	79.92
DESSN	2022	96.67	85.27	75.97	80.35	67.16	78.54
ChangeFormer	2022	96.70	86.51	74.93	80.30	67.09	78.51
AMTNet	2023	98.07	89.76	88.66	89.21	80.51	88.15
USSFC	2023	96.65	85.28	75.68	80.20	66.94	78.37
EATDer	2024	97.30	84.90	85.06	84.98	73.89	83.50
SFEARNet		**98.17**	**90.25**	**89.30**	**89.77**	**81.44**	**88.77**

TABLE IV
QUANTITATIVE RESULTS (%) FOR THE CLCD DATASET. BOLD VALUES ARE BEST

Methods	Year	OA	Pre	Rec	F1	IOU	Kappa
FC-EF	2018	94.37	72.90	38.74	50.60	33.87	47.91
FC-Siam-Conc	2018	94.33	67.52	45.77	54.56	37.51	51.65
FC-Siam-Diff	2018	94.45	68.68	46.65	55.56	38.47	52.71
STANet	2020	95.07	68.75	61.97	65.18	48.35	62.54
BIT	2021	96.34	76.18	73.88	75.01	60.01	73.03
MSCANet	2022	96.26	76.89	71.01	73.84	58.52	71.82
DESSN	2022	95.25	70.07	63.05	66.37	49.67	63.82
ChangeFormer	2022	94.27	61.74	60.46	61.10	43.99	58.00
AMTNet	2023	96.68	**81.05**	72.34	76.45	61.87	74.67
USSFC	2023	95.50	69.77	69.84	69.81	53.62	67.38
EATDer	2024	95.06	65.15	72.25	68.52	52.11	65.84
SFEARNet		**96.77**	79.61	**76.15**	**77.85**	**63.73**	**76.11**

A quantitative presentation is provided for the SFEARNet method and existing methods in the performance of GZ-CD. The results can be found in Table III. The results demonstrate that SFEARNet attains the highest scores on all six principal indicators. Specifically, SFEARNet improves the IOU by 0.93, 7.56, and 9.17 percentage points compared to AMTNet, EATDer, and STANet. Moreover, SFEARNet outperforms AMTNet and EATDer by 0.57 and 4.79 percentage points, respectively, in terms of $F1$. This article presents a comparative analysis of SFEARNet and alternative methods using the GZ-CD dataset. This is illustrated in Fig. 7. In refining the edges of large change areas, it can be seen that SFEARNet shows remarkable performance. For regions with irregular boundaries, SFEARNet also demonstrates higher recognition accuracy. Although there are some missed detections in medium- and small-scale change areas, SFEARNet can still accurately identify the change regions and retain more detailed information.

This article compares the performance of SFEARNet on the CLCD dataset in both qualitative and quantitative terms. It can be seen in Table IV and Fig. 8. Of the quantified findings, SFEARNet scored highest on five measures: OA, Rec, $F1$, IOU, and Kappa. In the IOU metrics, SFEARNet has improved by 1.85 percentage points relative to AMTNet and 3.72 percentage points relative to BIT. SFEARNet at $F1$ improved by 1.40 percentage points over AMTNet, 2.84 percentage points over BIT, and 4.01 percentage points over MSCANet. The qualitative results show that SFEARNet achieves good recognition results in linear, small, large, and irregular regions. Although there are some missed detections, SFEARNet can still identify areas of change while retaining edge information.

SFEARNet works best on the datasets of LEVIR-CD, WHU-CD, GZ-CD, and CLCD. The results are evidence of the effectiveness and robustness of this approach. SFEARNet shows lower detection error and more accurate edge detection, both quantitatively and qualitatively. It clearly demonstrates

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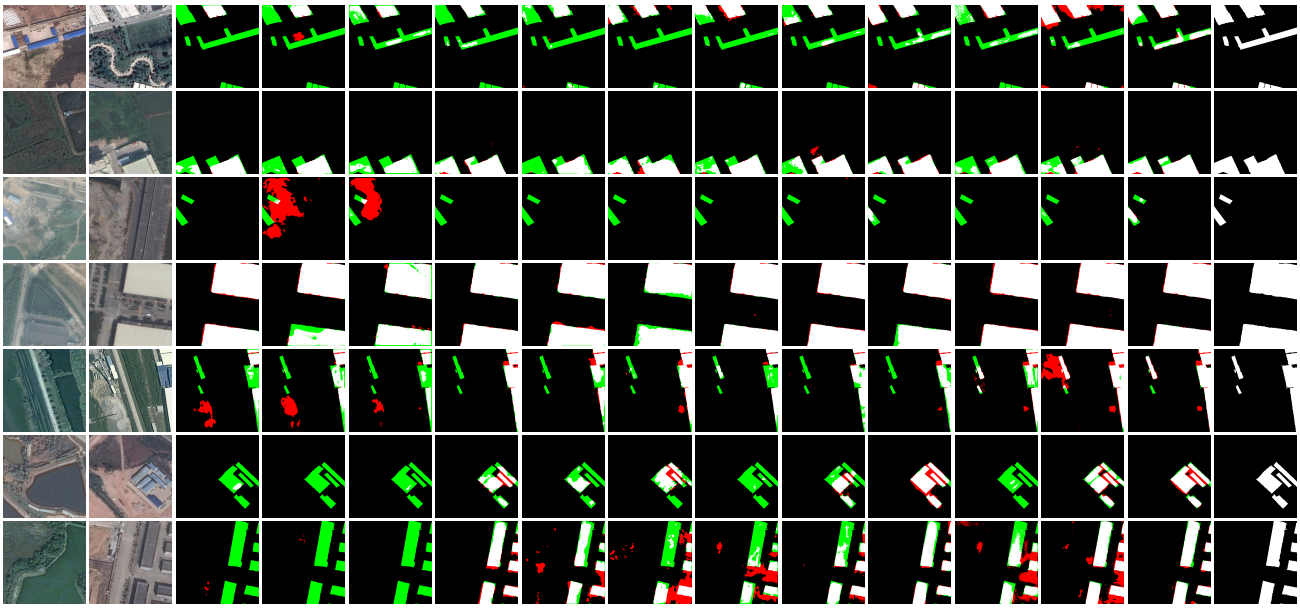


Fig. 7. Visualization comparisons on the GZ-CD dataset. (a) T1 image. (b) T2 image. (c) FC-EF. (d) FC-Siam-Conc. (e) FC-Siam-Diff. (f) STANet. (g) BIT. (h) MSCANet. (i) DESSN. (j) ChangeFormer. (k) AMTNet. (l) USSFC. (m) EATDer. (n) SFEARNet. (o) Ground truth. White indicates true positive. Black indicates true negative. Red indicates false positive. Green indicates false negative.

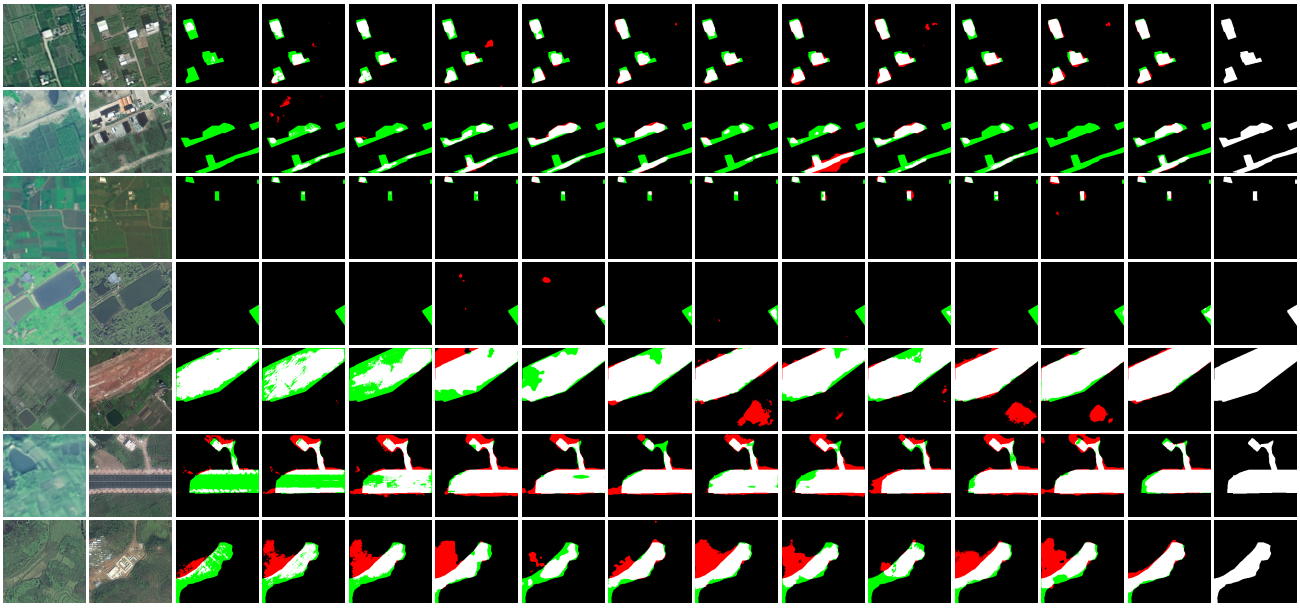


Fig. 8. Visualization comparisons on the CLCD dataset. (a) T1 image. (b) T2 image. (c) FC-EF. (d) FC-Siam-Conc. (e) FC-Siam-Diff. (f) STANet. (g) BIT. (h) MSCANet. (i) DESSN. (j) ChangeFormer. (k) AMTNet. (l) USSFC. (m) EATDer. (n) SFEARNet. (o) Ground truth. White indicates true positive. Black indicates true negative. Red indicates false positive. Green indicates false negative.

SFEARNet's superior performance in bitemporal change detection.

V. Discussion

This section will be a discussion of the effectiveness of SFITM, PFEM, EARM, and loss. The limitations of the proposed method will be explored.

A. Effectiveness of SFITM

SFITM improves network performance by enhancing the original upsampling method. This experiment is based on the SegFormer network, utilizing MixTransformer-b0 as the backbone network to construct a baseline network for comparison. Method-1 integrates SFITM into the baseline network. Method-1 is designed to evaluate the influence of SFITM. A comprehensive evaluation of the performance of SFITM was conducted through the utilization of four distinct datasets. This includes LEVIR-CD, WHU-CD, GZ-CD, and CLCD.

The outcomes of the quantitative experiments conducted on the four datasets are presented in Table V. The results show that SFITM led to significant improvements in a number of metrics. On the LEVIR-CD, Pre increased by 0.57 percentage

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TABLE V
QUANTITATIVE COMPARISON RESULTS (%) OF THE EFFECTIVENESS OF SFITM. BOLD VALUES ARE BEST

Dataset	Methods	SFITM	OA	Pre	Rec	F1	IOU	Kappa
LEVIR-CD	Baseline		98.91	90.13	88.19	89.15	80.43	88.58
	Method-1	√	**98.99**	**90.70**	**89.39**	**90.04**	**81.89**	**89.51**
WHU-CD	Baseline		99.29	93.99	88.64	91.24	83.88	90.87
	Method-1	√	**99.35**	**95.06**	**88.95**	**91.90**	**85.02**	**91.56**
GZ-CD	Baseline		97.83	**88.31**	87.40	87.85	78.34	86.66
	Method-1	√	**97.95**	87.50	**89.95**	**88.71**	**79.71**	**87.58**
CLCD	Baseline		96.08	74.32	72.26	73.28	57.82	71.16
	Method-1	√	**96.44**	**76.25**	**75.74**	**76.00**	**61.29**	**74.07**

points, while IOU increased by 1.46 percentage points. The WHU-CD showed an increase of 1.13 percentage points in the IOU and 1.06 percentage points in the Pre. In the GZ-CD, IOU increased by 1.37 percentage points and Rec by 2.55 percentage points. The CLCD dataset demonstrated a 3.48 percentage point increase in Rec and a 3.46 percentage point increase in IOU. The effectiveness of SFITM is demonstrated by the quantitative results presented above. As illustrated in Fig. 9, Method-1 demonstrated a substantial enhancement in comparison to the baseline. The incorporation of the SFITM enables the model to discern alterations in the images with greater precision. This observation is in accordance with the quantitative results, thereby further substantiating the benefits of SFITM in enhancing network performance.

In terms of $F1$, IOU, Kappa, and recall, SFITM has shown significant advantages. The comparative analysis of LEVIR-CD, WHU-CD, GZ-CD, and CLCD confirms this. The efficacy of SFITM in optimizing network performance is confirmed by the results.

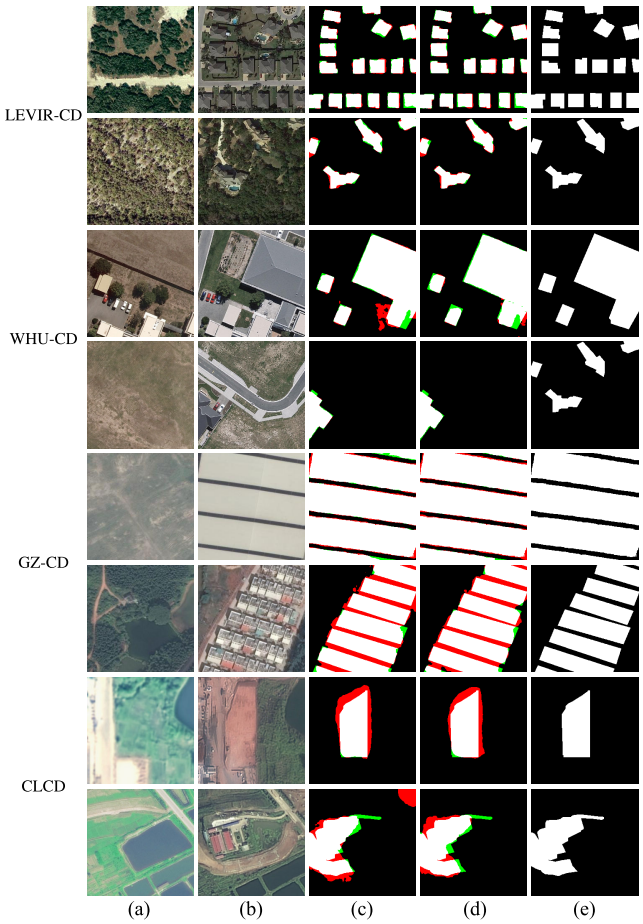


Fig. 9. Visualization comparisons of the effectiveness of SFITM. (a) T1 image. (b) T2 image. (c) Baseline. (d) Method-1. (e) Ground truth. White indicates true positive. Black indicates true negative. Red indicates false positive. Green indicates false negative.

B. Effectiveness of PFEM and EARM

PFEM improves features by combining different characteristics. EARM focuses on extracting and refining modified edge information.

It is essential to note that the edge information utilized by EARM is derived from the feature values generated by PFEM. The corrected edge information will affect the feature values generated by PFEM, thereby effectively combining the two modules. Therefore, this article evaluates the effectiveness of PFEM and EARM together. Four different datasets (LEVIR-CD, WHU-CD, GZ-CD, and CLCD) were used for the experiments. These experiments will comprehensively evaluate the performance of PFEM and EARM. PFEM was added to Method-1 to construct the new Method-2. SFEARNet comprehensively applies SFITM, PFEM, and EARM.

The experimental findings indicate that the model's performance has been enhanced with PFEM and EARM. It is illustrated in Table VI. The introduction of PFEM increased the IOU by 0.53 percentage points and the $F1$ by 0.32 percentage points on LEVIR-CD. The introduction of EARM resulted in an increase of 0.82 percentage points in IOU and 0.49 percentage points in $F1$, respectively. The application of PFEM to the WHU-CD dataset was observed to result in an increase in IOU by 0.37 percentage points and in Rec by 2.44 percentage points. The introduction of EARM led to a further increase in IOU and Pre by 0.42 percentage points and 1.40 percentage points, respectively, thereby confirming the efficacy of EARM. In the GZ-CD dataset, while PFEM exhibited a 0.90 percentage point improvement in IOU, Pre demonstrated a 2.76 percentage point increase. The introduction of EARM has been shown to increase Rec by 1.02 percentage points and IOU by 0.83 percentage points. The following results were obtained from the CLCD dataset. The outcomes lead to a 0.47 percentage point enhancement in $F1$ and a 0.62 percentage point growth in IOU with the integration of the PFEM module. The incorporation of EARM

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TABLE VI
QUANTITATIVE COMPARISON RESULTS (%) OF THE EFFECTIVENESS OF PFEM AND EARM. BOLD VALUES ARE BEST

Dataset	Methods	SFITM	PFEM	EARM	OA	Pre	Rec	F1	IOU	Kappa
LEVIR-CD	Method-1	√			98.99	90.70	89.39	90.04	81.89	89.51
	Method-2	√	√		99.02	90.91	89.81	90.36	82.41	89.84
	SFEARNet	√	√	√	**99.07**	**91.43**	**90.27**	**90.85**	**83.23**	**90.36**
WHU-CD	Method-1	√			99.35	**95.06**	88.95	91.90	85.02	91.56
	Method-2	√	√		99.35	92.86	**91.39**	92.12	85.39	91.78
	SFEARNet	√	√	√	**99.38**	94.25	90.55	**92.36**	**85.81**	**92.04**
GZ-CD	Method-1	√			97.95	87.50	**89.95**	88.71	79.71	87.58
	Method-2	√	√		98.09	**90.27**	88.28	89.26	80.61	88.22
	SFEARNet	√	√	√	**98.17**	90.25	89.30	**89.77**	**81.44**	**88.77**
CLCD	Method-1	√			96.44	76.25	75.74	76.00	61.29	74.07
	Method-2	√	√		96.52	76.99	75.96	76.47	61.90	74.59
	SFEARNet	√	√	√	**96.77**	**79.61**	**76.15**	**77.85**	**63.73**	**76.11**

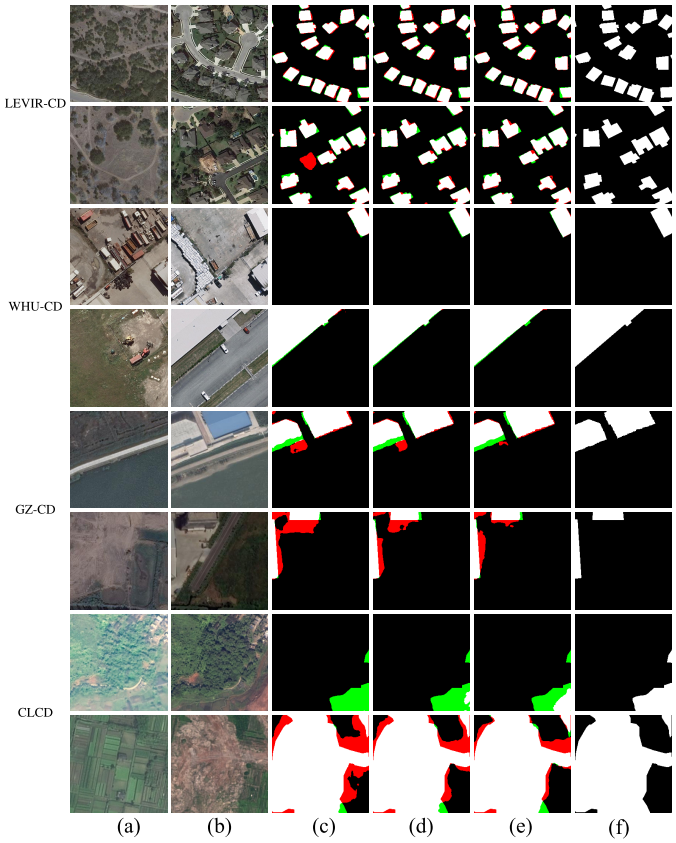


Fig. 10. Visualization comparisons of the effectiveness of PFEM and EARM. (a) T1 image. (b) T2 image. (c) Method-1. (d) Method-2. (e) SFEARNet. (f) Ground truth. White indicates true positive. Black indicates true negative. Red indicates false positive. Green indicates false negative.

resulted in an enhancement in IOU by 1.83 percentage points and Pre by 2.63 percentage points. This evidence serves to illustrate the beneficial impact of PFEM and EARM on the enhancement of change detection performance.

To intuitively demonstrate the experimental results, qualitative analysis was also conducted on each dataset. As shown in Fig. 10, compared to Method-1, our method with PFEM and EARM can more accurately identify changed regions, especially in the edge areas. This result is consistent with the quantitative analysis. PFEM and EARM are also shown to be effective at improving change detection accuracy.

In conclusion, this article presents a comprehensive demonstration of the benefits of PFEM and EARM in enhancing performance. It can be evidenced by both qualitative and quantitative evaluations. The introduction of these two modules not only improves the model's Pre, IOU, and Kappa but also enhances its robustness and generalization ability when dealing with complex scenes.

C. Effectiveness of Loss Function

CE loss, Jaccard loss, and dice loss are all commonly utilized loss functions in the field of deep learning. By using the three loss functions together, the performance of the model can be improved.

This article sets up ablation experiments to verify the effectiveness of each loss function. Method-3 is the network model using only CE loss. Method-4 is the network model using both CE loss and Jaccard loss. SFEARNet uses all three loss functions simultaneously.

The efficacy of the loss function can be demonstrated in Table VII. It is clear that adding Jaccard loss results in better accuracy for all the datasets. There was an increase of 0.54 percentage points in the IOU for the LEVIR-CD dataset and an increase of 0.34 percentage points in the Rec dataset. On the WHU-CD dataset, the IOU and pre metrics showed an improvement of 0.42 and 0.72 percentage points. The GZ-CD dataset shows an IOU increase of 2.18 percentage points and a pre increase of 2.11 percentage points. On the CLCD dataset, IOU and Kappa increased by 2.07 and 1.71 percentage points. Dice-loss similarly improves model performance. For the LEVIR-CD, WHU-CD, GZ-CD, and CLCD datasets, the IOU increases by 0.54, 0.59, 1.63, and 0.19 percentage points. $F1$ increases by 0.57, 0.34, 1.00, and 0.14 percentage points. To provide a more visual demonstration of each loss function, the results of each dataset are presented visually in Fig. 11. Method-4 and SFEARNet are better than Method-3. This finding serves to substantiate the efficacy of the individual loss functions.

This article demonstrates the advantages of composite loss function. The Rec, $F1$, IOU, and Kappa of the model have been shown to be improved by the composite loss function. Moreover, the model's stability and robustness are enhanced.

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TABLE VII
QUANTITATIVE COMPARISON RESULTS (%) OF THE EFFECTIVENESS OF LOSS FUNCTION. BOLD VALUES ARE BEST

Dataset	Methods	CE-Loss	Jaccard-Loss	Dice-Loss	OA	Pre	Rec	F1	IOU	Kappa
LEVIR-CD	Method-3	√			98.99	91.45	88.51	89.95	81.74	89.42
	Method-4	√	√		99.03	**91.76**	88.85	90.28	82.28	89.77
	SFEARNet	√	√	√	**99.07**	91.43	**90.27**	**90.85**	**83.23**	**90.36**
WHU-CD	Method-3	√			99.32	92.92	**90.66**	91.77	84.80	91.42
	Method-4	√	√		99.35	93.64	90.46	92.02	85.22	91.68
	SFEARNet	√	√	√	**99.38**	**94.25**	90.55	**92.36**	**85.81**	**92.04**
GZ-CD	Method-3	√			97.79	89.23	85.66	87.41	77.64	86.20
	Method-4	√	√		98.04	**91.34**	86.35	88.77	79.81	87.70
	SFEARNet	√	√	√	**98.17**	90.25	**89.30**	**89.77**	**81.44**	**88.77**
CLCD	Method-3	√			96.56	78.68	73.75	76.14	61.47	74.29
	Method-4	√	√		**96.82**	**81.12**	74.57	77.71	63.54	75.99
	SFEARNet	√	√	√	96.77	79.61	**76.15**	**77.85**	**63.73**	**76.11**

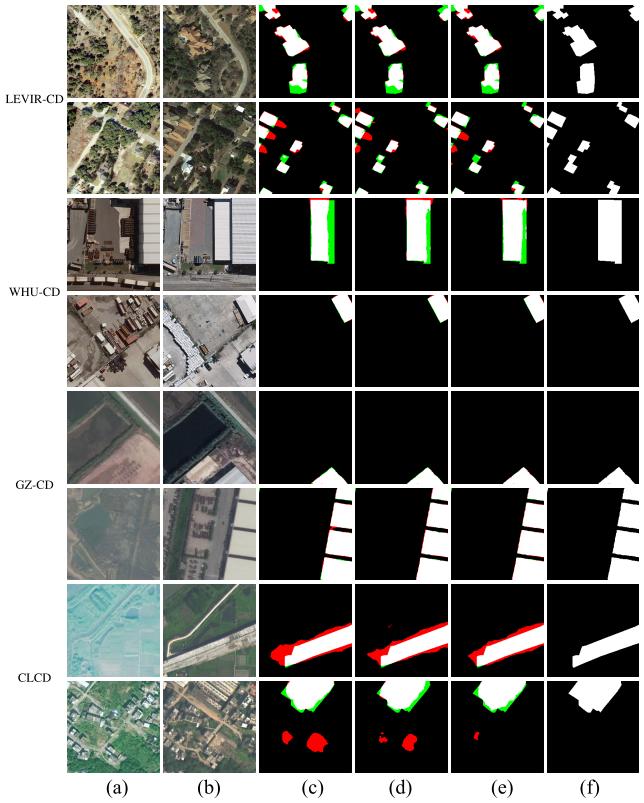


Fig. 11. Visualization comparisons of the effectiveness of loss function. (a) T1 image. (b) T2 image. (c) Method-3. (d) Method-4. (e) SFEARNet. (f) Ground truth. White indicates true positive. Black indicates true negative. Red indicates false positive. Green indicates false negative.

D. Efficiency Comparison

The number of Params and floating-point operations per second (FLOPs) are used to evaluate the performance of SFEARNet. This evaluation validates SFRAENet's effectiveness and efficiency. The complexity of the model is indicated by the number of Params. In contrast, the concept of FLOPs is a parameter for evaluating the time complexity of the model.

The Params and the FLOPs of the proposed SFEARNet are presented in comparison with other methods. It is given in Table VIII. The comparison unambiguously illustrates the benefits of SFEARNet in relation to Params and FLOPs. SFEARNet has the highest $F1$ and IOU across the different datasets although it has a moderate level of Params. This indicates that SFEARNet has a low level of spatial complexity while achieving excellent performance in change detection. SFEARNet shows a reduction in the number of Params while achieving a reduction in the number of FLOPs compared to DESSN and ANTNet. These findings indicate that SFEARNet not only reduces storage space demands but also enhances computational efficiency, which is of significant importance in constrained computing environments. This evidence indicates that SFEARNet is able to maintain high accuracy while achieving the lowest time complexity. This makes it an ideal choice for applications requiring fast response times.

In summary, there are significant efficiency and performance advantages to the proposed SFEARNet network. SFEARNet outperforms other methodologies on all datasets and, thus, demonstrates its broad applicability and robustness in diverse scenarios. In addition, SFEARNet demonstrates excellent performance and the lowest FLOPs among models with comparable parameter counts. This further substantiates the equilibrium of SFEARNet in regard to model complexity, precision, and computational efficacy.

E. Limitations of SFEARNet

The aforementioned method, SFEARNet, has demonstrated an acceptable equilibrium between efficiency and accuracy in change detection tasks. However, it must be acknowledged that certain inherent constraints do exist, which require further attention in subsequent research. The involvement of complex matrix operations and the intricate network architecture results in a relatively high parameter size for SFEARNet. The high parameter size necessitates a greater demand for computational resources and storage, which may impede the deployment of SFEARNet in resource-constrained environments. Despite SFEARNet's remarkable impact on edge extraction for changing areas and its ability to capture more accurate and refined edge features, its performance is still limited by the complexity of the network and the variety of feature types. This constraint suggests that SFEARNet might continue to encounter instances of either missed or false detection in particular contexts. The existence of these phenomena indicates that there is still scope for further optimization of the proposed method. Future research may benefit from

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TABLE VIII
COMPARISON RESULTS OF EFFICIENCY OF DIFFERENT METHODS. BOLD VALUES ARE BEST

Methods	Year	Params(M)	FLOPs(G)	LEVIR-CD		WHU-CD		GZ-CD		CLCD	
				F1(%)	IOU(%)	F1(%)	IOU(%)	F1(%)	IOU(%)	F1(%)	IOU(%)
FC-EF	2018	1.35	3.58	75.40	60.51	74.26	59.06	65.67	48.89	50.60	33.87
FC-Siam-Conc	2018	1.55	5.33	86.08	75.57	69.06	52.74	63.69	46.72	54.56	37.51
FC-Siam-Diff	2018	1.35	4.73	85.84	75.20	76.37	61.78	60.73	43.60	55.56	38.47
STANet	2020	16.93	41.72	89.19	80.49	87.86	78.34	83.91	72.28	65.18	48.35
BIT	2021	3.04	8.75	89.60	81.16	90.65	82.89	81.87	69.31	75.01	60.01
MSCANet	2022	16.42	14.80	89.73	81.37	90.59	82.80	81.60	68.92	73.84	58.52
DESSN	2022	34.64	69.53	88.96	80.11	86.72	76.55	80.35	67.16	66.37	49.67
ChangeFormer	2022	41.03	202.79	89.73	81.36	83.78	72.09	80.30	67.09	61.10	43.99
AMTNet	2023	24.67	21.56	90.50	82.66	91.94	85.09	89.21	80.51	76.45	61.87
USSFC	2023	1.52	4.86	88.82	79.89	87.52	77.81	80.20	66.94	69.81	53.62
EATDer	2024	6.61	23.46	89.91	81.68	86.60	76.36	84.98	73.89	68.52	52.11
SFEARNet		5.56	4.65	**90.85**	**83.23**	**92.36**	**85.81**	**89.77**	**81.44**	**77.85**	**63.73**

investigating techniques that could improve the performance of the aforementioned network, such as model pruning, quantization, or knowledge distillation. These methods can reduce the parameter size and computational requirements while maintaining or even improving accuracy. Furthermore, an investigation into more sophisticated network architectures or feature fusion methodologies could assist SFEARNet in more effectively processing intricate feature types, thereby reducing the occurrence of missed and false detections.

VI. CONCLUSION

This article presents a network combining semantic flow and edge-aware refinement for highly efficient RS image change detection named SFEARNet. The network has been constructed based on the SegFormer architecture and integrates three key modules. These are the PFEM, the SFITM, and the EARM. PFEM strengthens the differential information of change areas. SFITM is designed to minimize information loss during the upsampling. EARM is designed to refine the edge of the changed regions. It makes edges more precise and refined. The collective impact of these modules has a discernible positive influence. They also excel in the precision and comprehensiveness of altered region delineation.

This article uses a combination of multiscale features. PFEM combines different scale images at different dilatation rates. The module combines the original differential, contextual, and attention features. In SFITM, the transfer of semantic information between deep and shallow features is employed to obtain high-resolution feature maps with high semantic information content. The extensive use of multiscale features enhances the network's ability to capture rich detail and high-level semantics in the image. It also improves its stability and adaptability in the face of inputs of different scales and complexities.

In this article, fine edge feature generation and correction are applied to obtain more accurate and detailed edge information in EARM using the dice loss function. This edge information is further used to make a refinement to the extracted features. The edge region of the feature map is adjusted and optimized in order to make the final obtained feature values more explicit and clearer at the edge. The inclusion of EARM makes the network show a high degree of accuracy and robustness.

To assess the efficacy of SFEARNet, comparative experiments were conducted on the LEVIR-CD, WHU-CD, GZ-CD, and CLCD datasets. The findings of the experiments were categorized according to three distinct analytical approaches: qualitative, quantitative, and network efficiency analyses. These allow for the verification of the robustness and generalizability of SFEARNet. The results further substantiate the effect of SFITM, PFEM, and EARM in enhancing model performance. This article also illustrates the efficacy of mixed losses. SFEARNet achieves an optimal balance between efficiency and accuracy. In particular, in networks with the same parameter scale, SFEARNet demonstrated the lowest FLOPs and achieved the highest efficiency. SFEARNet exhibited slightly superior accuracy and demonstrated a notable enhancement in efficiency.

Although SFEARNet has reached a certain equilibrium between efficiency and accuracy, further enhancements are still necessary with regard to parameter size. Moreover, in scenarios characterized by complexity, the process of change detection can be significantly influenced by a number of external factors, including lighting and shadows. In future research, the objective will be to enhance the model's resilience to interference, reduce the model's parameter size, and accelerate the model's inference speed. Furthermore, additional research will be undertaken to investigate the viability of cross-device change detection solutions in environments with limited resources.

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